

Safe real-time optimization of a distillation column under feed-composition uncertainty: a comparative study of distribution-free constraint back-offs

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Abstract

Real-time optimization (RTO) pushes a distillation column toward its product specifications, so the optimizer must subtract a back-off from any specification it cannot measure online. Data-driven back-offs sized with conformal prediction are attractive because they need no disturbance model, but their guarantee holds on average over operating points, and the optimizer does not choose operating points at random. This study compares six ways of sizing the back-off of a chance-constrained RTO for a rigorous DWSIM Peng-Robinson debutanizer twin under an unmeasured feed-composition disturbance, scoring every deployed setpoint by its exact violation probability over 1000 repeated trials. At the realistic disturbance level, conformal back-offs that are valid on average deploy a setpoint whose violation probability exceeds the 10% target in 67.5% to 100% of trials, and an a posteriori plug-in correction still does so in 32.0%. Certify-then-deploy, a fixed-sequence binomial test of candidate setpoints, and scenario-based sampling-and-discarding keep this rate at or below 2.5% while staying within 0.06% to 0.21% of the oracle profit. A risk-adjusted analysis shows that the unguaranteed back-offs lose money in expectation once off-spec bottoms cost more than 0.017 to 0.032 USD/kg, about 2% to 4% of the product value. The results give a selection rule: use sampling-and-discarding when the plant model fits inside the optimizer, and certify-then-deploy when it does not.

Keywords: chance constraint, conformal prediction, digital twin, distillation, real-time optimization, scenario approach

1 INTRODUCTION

Real-time optimization (RTO) recomputes the setpoints of a process unit as prices, feeds, and equipment drift, and it earns its value by operating closer to constraints than a fixed recipe would. In a debutanizer the most profitable operation keeps as much material as possible in the more valuable stream, which drives the bottoms composition toward its specification. That specification is usually not measured online. It is known through laboratory analyses or an inferential estimate, so the optimizer must subtract a back-off from the specification to protect against disturbances it cannot see. The back-off is the central design quantity of RTO under uncertainty: too small

and the column produces off-spec product whenever the disturbance is adverse; too large and the economic benefit of the optimization is given away. Adaptation schemes correct the optimizer for plant-model mismatch,^[1,2] and data-driven models of distillation columns have recently been placed inside RTO loops,^[3] but both still need a principled rule for sizing the margin on an unmeasured quality.

The classical rule is a chance constraint, which bounds the probability of violating the specification under an assumed disturbance distribution.^[4] Distribution-free alternatives remove that assumption. Conformal prediction turns any predictor into an interval whose coverage holds in finite samples when the data are exchangeable, without a model of the disturbance^[5–7]; conformalized quantile regression adapts the interval width to the operating point.^[8] Placing the upper edge of such an interval as the back-off is therefore a natural way to size the margin from plant or simulator data, and conformal constructions have recently been proposed inside optimization and control.^[9,10]

Back-offs themselves are well established in process systems engineering. Stochastic back-off algorithms size margins for integrated design, control, and scheduling under uncertainty^[11,12]; explicit back-offs tighten the constraints of stochastic nonlinear model predictive control^[13,14]; Gaussian-process models supply back-offs in data-driven predictive control^[15]; and back-offs can be tuned by Bayesian optimization from the empirical distribution of constraint values.^[16] Each of these studies proposes and evaluates one construction. Conformal prediction has also been linked to scenario optimization at the level of guarantees,^[17,18] and conformal constraint learning has been proposed for mixed-integer design.^[19]

The difficulty is that conformal coverage is a statement about an average. It holds over the distribution of operating points used for calibration, while an optimizer deliberately seeks the boundary of the feasible region, where the margin is tightest relative to the local risk. Selecting the best point by optimization favours exactly the points where the margin is optimistic, an instance of the optimizer’s curse known in decision analysis,^[20] and coverage conditional on the chosen operating point cannot be guaranteed without further structure.^[21] Remedies exist in the statistics and control literature: calibrating the chosen solution again on an independent sample,^[9] tuning a margin by fixed-sequence testing,^[22,23] and the scenario approach with its sampling-and-discarding variant.^[24–26] What a process engineer lacks is a like-for-like comparison of these options on a rigorous process model, with the consequences stated in the units that matter in a plant: how often an unsafe setpoint is actually deployed, what the guarantee costs in profit, and which option to use in which situation.

This paper supplies that comparison. No new method is proposed, and the contribution is stated accordingly. To the author’s knowledge, this is the first controlled head-to-head comparison of distribution-free back-off constructions (split conformal, conformalized quantile regression, an

a posteriori plug-in correction, fixed-sequence certify-then-deploy, the scenario approach, and sampling-and-discarding) for chance-constrained steady-state RTO of a distillation column. The novel elements are (i) exact violation scoring of every deployed setpoint over 1000 repeated trials on a rigorous DWSIM Peng-Robinson twin, (ii) the rate at which each construction deploys an unsafe setpoint across a range of disturbance magnitudes, (iii) a risk-adjusted economic analysis that converts the safety differences into a breakeven off-spec cost, and (iv) a selection rule and a certification-budget table that can be applied before commissioning. Section 2 states the problem, Section 3 the constructions compared, Section 4 the twin and the experiment, Section 5 the results, and Section 6 the selection guide and limitations.

2 PROBLEM FORMULATION

The decision is the operating point $u = (R, D)$, with R the reflux ratio and D the distillate flow in kmol/h, restricted to the box $0.8 \leq R \leq 3.0$, $33 \leq D \leq 37$. The disturbance z is the n -butane mole fraction of the feed. It is neither manipulated nor measured online, and its distribution P is unknown to the optimizer; successive values are treated as independent draws over the RTO time scale. The constrained quality is the bottoms n -butane mole fraction $g(u, z) = x_B$, with specification $x_B \leq \bar{g} = 0.02$. The operating profit $\pi(u)$ values the overhead at the liquefied petroleum gas price and the bottoms at the stabilized-gasoline price, both by mass, less the reboiler energy cost.

The chance-constrained RTO is formulated as follows:

$$\max_u \pi(u) \quad \text{s.t.} \quad P(g(u, z) \leq \bar{g}) \geq 1 - \alpha, \quad (1)$$

with $\alpha = 0.10$. Writing $V(u) = P(g(u, z) > \bar{g})$ for the violation probability of a setpoint, the constraint reads $V(u) \leq \alpha$. Because V is not available, the optimizer uses a nominal model $\hat{g}(u) = g(u, z_0)$ at the nominal composition $z_0 = 0.35$ and a back-off $C(u) \geq 0$ as follows:

$$\max_u \pi(u) \quad \text{s.t.} \quad \hat{g}(u) + C(u) \leq \bar{g}. \quad (2)$$

Two kinds of guarantee must be distinguished. A back-off is *marginally valid* if the probability that g exceeds $\hat{g} + C$ is at most α on average over the operating points used to calibrate it. The engineering requirement is different: the plant runs at the one setpoint the optimizer deploys, so what matters is whether that setpoint satisfies $V \leq \alpha$. This paper therefore judges every method by the deployed-decision criterion as follows:

$$P(V(u_{\text{dep}}) > \alpha) \leq \delta, \quad (3)$$

99 with $\delta = 0.05$, where the probability is over the data used to build and check the back-off.

100 3 BACK-OFF CONSTRUCTIONS COMPARED

101 3.1 Marginal conformal back-offs

102 A calibration set $\{(u_i, z_i, g_i)\}_{i=1}^n$ of evaluations at random operating points gives residuals
 103 $r_i = g_i - \hat{g}(u_i)$. The *fixed* back-off is the one-sided split-conformal quantile of the residuals,
 104 $C = r_{(q)}$ with $q = \lceil (n+1)(1-\alpha) \rceil$ ^[5,6]; it spends the same margin everywhere. The *conformalized*
 105 *quantile regression* (CQR) back-off fits a gradient-boosted estimate $\hat{q}(u)$ of the conditional
 106 $(1-\alpha)$ quantile of g and adds the conformal quantile of $g_i - \hat{q}(u_i)$,^[8] so its width follows the
 107 local risk. Both are marginally valid; neither is guaranteed at the setpoint the optimizer selects.

108 3.2 A posteriori plug-in correction

109 A common repair solves Equation (2) with the CQR back-off, then inflates it by the smallest
 110 factor $\kappa \geq 1$ whose resulting setpoint meets the target on a validation sample of disturbances,
 111 in the spirit of the a posteriori calibration of conformal predictive programming.^[9] In the
 112 implementation compared here the validation sample has 6000 draws and a tolerance of 0.01 is
 113 allowed. The correction is accurate on average, but κ is chosen with the same sample that judges
 114 it, so it carries no guarantee of the form Equation (3).

115 3.3 Certify-then-deploy

116 Certify-then-deploy separates choosing candidates from checking them. For a decreasing list
 117 $\kappa_1 > \kappa_2 > \dots > \kappa_J$ fixed in advance, the optimizer produces candidate setpoints u_j by solving
 118 Equation (2) with $\kappa_j C_{\text{CQR}}$. An independent certification sample $z'_1, \dots, z'_m$ is then drawn and each
 119 candidate is evaluated on it, giving the violation count N_j . The one-sided Clopper-Pearson upper
 120 bound U_j on $V(u_j)$ at confidence $1 - \delta$ is computed from N_j .^[27] Candidates are tested in order,
 121 from the most conservative, and testing stops at the first candidate with $U_j > \alpha$; the last candidate
 122 that passed is deployed, and if none passes the plant stays at a conservative fallback setpoint.

123 **Proposition 1.** *If the certification sample is independent of everything used to produce the*
 124 *candidates, then $P(\text{a setpoint with } V > \alpha \text{ is deployed}) \leq \delta$, whatever the model, optimizer, or*
 125 *back-off used to produce the candidates.*

126 *Proof.* Conditional on the candidates, N_j is binomial with parameters m and $V(u_j)$, so
 127 $P(U_j < V(u_j)) \leq \delta$ for each j . Let $j^\dagger$ be the first candidate with $V(u_{j^\dagger}) > \alpha$. Because testing

stops at the first failure, an unsafe candidate can be deployed only if candidate $j^\dagger$ passes, that is, only if $U_{j^\dagger} \leq \alpha < V(u_{j^\dagger})$, which has probability at most δ . $\square$

This is the fixed-sequence argument used for risk control in Learn-then-Test.^[22] It needs no multiplicity correction and no convexity, and the sample size m does not depend on the number of decision variables. It does require that the plant model g can be evaluated at the candidate setpoints, which is the case for a validated simulator or for pilot runs at the candidate conditions.

3.4 Scenario approach and sampling-and-discarding

The scenario approach imposes $g(u, z_i) \leq \bar{g}$ for N sampled disturbances. With two decision variables, $N = 46$ gives $P(V > 0.10) \leq 0.05$.^[24,25] Sampling-and-discarding draws more scenarios and removes the k most restrictive ones while keeping the same guarantee^[26]; with $N = 1000$, $k = 69$. Both evaluate the plant model inside the optimizer.

3.5 Oracle

The oracle knows the disturbance distribution, sets $C(u) = g(u, z_{1-\alpha}) - \hat{g}(u)$ with $z_{1-\alpha}$ the $(1 - \alpha)$ quantile of z (valid because x_B increases with z), and serves only as the reference.

4 CASE STUDY AND EXPERIMENTAL DESIGN

4.1 Digital twin

The plant is a DWSIM 9.0.5^[28] model of a debutanizer reduced to the binary system n -butane/ n -hexane under the Peng-Robinson equation of state^[29]: nine equilibrium stages with the condenser as stage 0 and the reboiler as stage 8, feed on stage 4, top pressure 470 kPa (4.7 bar) with a 40 kPa (0.4 bar) column pressure drop, and a feed of 100 kmol/h. A nominal campaign at $z_0 = 0.35$ covers a 4×4 grid of $R \in \{0.8, 1.5, 2.2, 3.0\}$ and $D \in \{33, 34.5, 36, 37\}$ kmol/h (15 converged runs), and a disturbance campaign repeats the grid at $z \in \{0.300, 0.325, 0.350, 0.375, 0.400\}$ (78 runs). Gaussian-process surrogates fitted to the nominal runs give $\hat{g}$, the distillate purity, and the reboiler duty; a Gaussian process over (R, D, z) fitted to the disturbance campaign serves as the truth g . Leaving out the whole $z = 0.375$ slice and predicting it gives $R^2 = 0.917$ and a mean absolute error of 0.0053 in x_B . Figure 1 shows the loop.

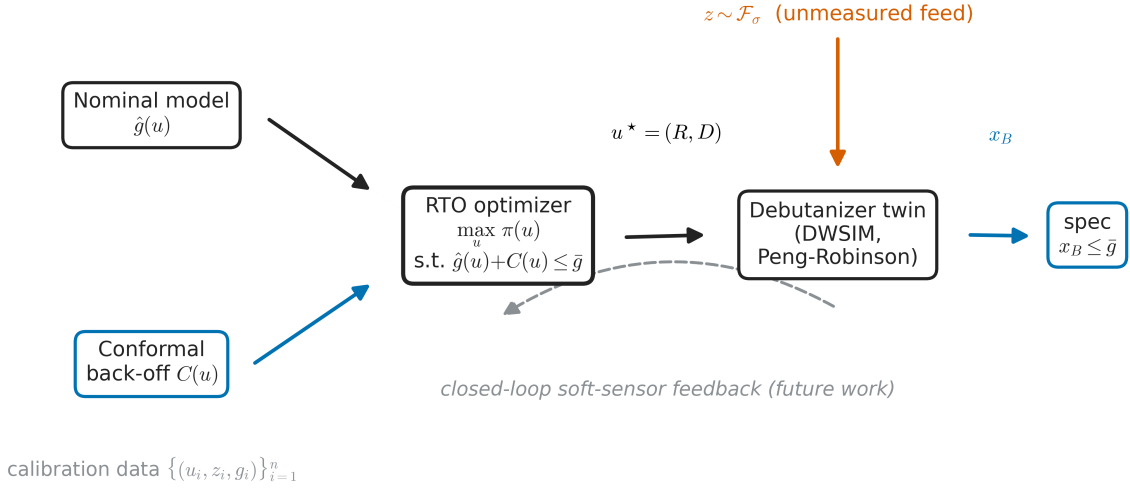


FIGURE 1 Chance-constrained real-time optimization (RTO) with a data-driven back-off. The optimizer uses the nominal model and the back-off; the feed composition z is not measured, and neither is the bottoms composition x_B it constrains.

4.2 Economics

Overhead is valued at the liquefied petroleum gas price (0.340 USD/kg) and bottoms at the stabilized-gasoline price (0.836 USD/kg), and reboiler energy costs 6.28 USD/GJ. Because the gasoline outvalues the light key, profit rises as the column lets more material leave with the bottoms, which pushes x_B toward its specification and makes the chance constraint economically active.

4.3 Experiment

The optimizer is an exact search over a 67×61 grid of (R, D) (4087 setpoints), so no method is advantaged by a local solver. The feed composition is a normal variable centred on 0.35 and truncated to the campaign range of 0.30 to 0.40, with standard deviation σ_z of 0.006, 0.010, 0.015, 0.020, or 0.025; 0.006 is the realistic level for a well-controlled feed. The calibration set has $n = 1500 \max(1, \sigma_z/0.006)$ points. Each method is run in 200 independent trials per σ_z (1000 trials in total), with certification budgets m of 200 or 1000. Because x_B increases with z at 99.6% of the grid setpoints, the violation probability of any setpoint follows exactly from the composition at which it crosses the specification, so every deployed setpoint is scored without Monte Carlo error.

For the economic comparison, the risk-adjusted profit rate of a setpoint is defined as follows:

$$J(u) = \pi(u) - \Delta \dot{m}_B V(u), \quad (4)$$

where Δ is the cost per kilogram of off-spec bottoms (downgrading or reprocessing) and $\dot{m}_B = 5636$ kg/h is the bottoms flow at the oracle setpoint. A trial in which a method abstains is charged the conservative fallback setpoint, the most profitable one with $V \leq 0.01$. A synthetic benchmark with an exactly known violation probability (two decision variables, a skewed disturbance, sensitivity that grows in the profitable direction, 400 trials) is used to isolate the mechanism behind the failure of marginal back-offs; the code for both studies is public. Portions of the analysis and plotting code were written with the assistance of a large language model (see Acknowledgements); the author reviewed and tested all code, and every reported number regenerates from the public scripts and fixed random seeds.

4.4 Use of generative artificial intelligence tools

The author used Claude (Anthropic; Claude Opus 5.5 and earlier versions), a large language model, to assist with drafting and editing the manuscript text and with writing and debugging parts of the Python analysis and plotting code. The author directed the study, reviewed and tested all outputs, verified every reported number against the regenerated evidence files, and takes full responsibility for the content. No generative AI tool was used to create or alter simulation data or results; the figures were produced by the author’s code from the simulation data.

5 RESULTS AND DISCUSSION

5.1 Why valid-on-average back-offs fail under optimization

In the synthetic benchmark the fixed, normalized, and CQR back-offs all reach their marginal target exactly (average miss 0.100), yet the setpoint the optimizer selects is unsafe in 93% to 100% of trials, with a violation probability of 0.17 to 0.18. The cause is selection. The normalized back-off scales the margin by a local estimate of the residual spread; when that estimate is made noisier (fewer neighbours in a nearest-neighbour estimate), unsafe deployments rise from 61.0% to 99.7%, and even the smoothest estimate stays far above δ . An estimate that is unbiased everywhere is still optimistic somewhere, and the optimizer finds that place. In the same benchmark certify-then-deploy deploys an unsafe setpoint in 1.5% of trials (95% interval 0.6% to 3.2%), consistent with the proposition.

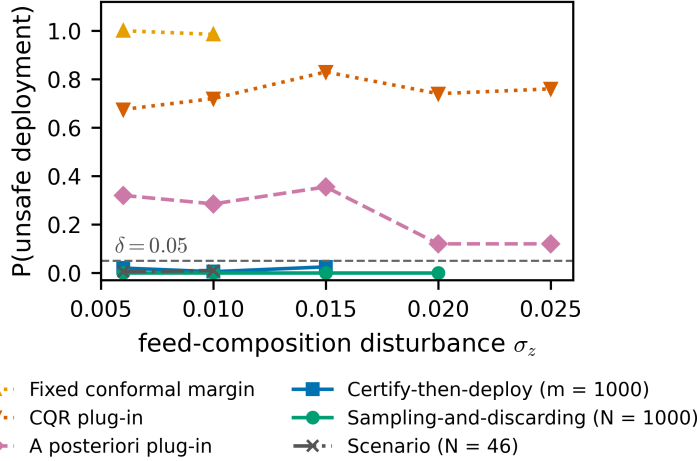


FIGURE 2 Fraction of trials in which the deployed setpoint violates the specification with probability above $\alpha = 0.10$, against the disturbance magnitude (200 trials per point; points shown where the method deploys in at least half of the trials). The dashed line marks $\delta = 0.05$.

TABLE 1 Twin results at the realistic disturbance ($\sigma_z = 0.006$, 200 trials).

Method	Unsafe deployments (%)	95% CI (%)	Deploys (%)	Δ profit (USD/h)	Mean V
Fixed conformal margin	100.0	98.2 to 100	100	+7.5	0.157
CQR plug-in	67.5	60.5 to 73.9	100	+4.5	0.147
A posteriori plug-in	32.0	25.6 to 38.9	93	-4.1	0.073
Certify-then-deploy, $m = 1000$	2.0	0.5 to 5.0	88	-11.5	0.051
Sampling-and-discarding, $N = 1000$	0.0	0 to 1.8	100	-3.2	0.064
Scenario, $N = 46$	0.5	0 to 2.8	100	-27.3	0.020

Note: Δ profit is the expected profit minus the oracle profit (5373.9 USD/h, $V = 0.083$, below 0.10 because of the finite grid); expected values include the fallback setpoint for abstaining trials; CI, confidence interval; CQR, conformalized quantile regression.

5.2 Safety on the distillation twin

Figure 2 and Table 1 give the twin results. At the realistic disturbance ($\sigma_z = 0.006$), the fixed conformal back-off deploys an unsafe setpoint in every trial and the CQR back-off in 67.5% of trials, with expected violation probabilities of 0.157 and 0.147 against a target of 0.10. The a posteriori plug-in correction lowers the expected violation to 0.073, close to the target on average, but still deploys an unsafe setpoint in 32.0% of trials; across all disturbance levels this rate ranges from 12.0% to 35.5%. An average that looks right is not a guarantee. Certify-then-deploy keeps unsafe deployments at or below 2.5% at every disturbance level and both budgets, and sampling-and-discarding at or below 0.5%.

When the plant model fits inside the optimizer, sampling-and-discarding is the stronger method on this problem: it is at least as safe as certify-then-deploy, it earns more (3.2 against 11.5 USD/h below the oracle), and it never abstains for $\sigma_z \leq 0.015$, whereas certify-then-deploy abstains

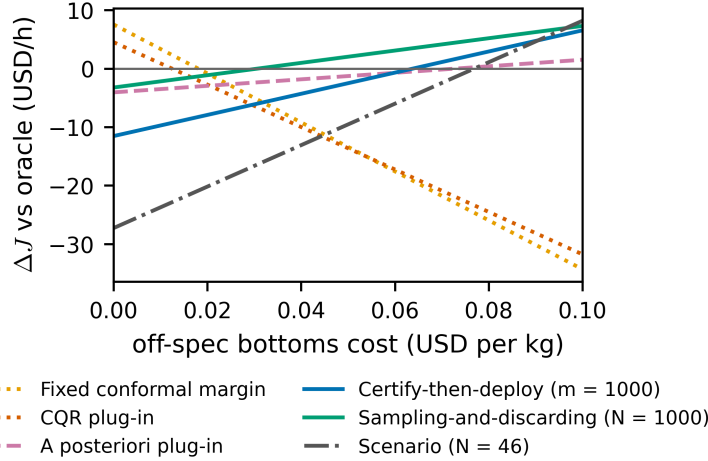


FIGURE 3 Expected risk-adjusted profit relative to the oracle, ΔJ , against the cost of off-spec bottoms at $\sigma_z = 0.006$. Crossings mark the breakeven costs quoted in the text.

in 12% of trials at the realistic level. With a single disturbance and a monotone response only one sampled disturbance limits the solution, so discarding extracts almost all the information in the samples. The basic scenario program with $N = 46$ is safe but gives up 27.3 USD/h. All the methods that meet Equation (3) stay within 0.06% to 0.21% of the oracle profit; the differences that matter are in safety, not in profit. Although conformal prediction and scenario optimization are linked at the level of their guarantees,^[17] the constructions compared here certify different quantities: the marginal back-offs certify an average over calibration setpoints, whereas certify-then-deploy and the scenario methods certify the setpoint that is deployed.

5.3 Risk-adjusted economics

The unguaranteed back-offs appear more profitable than the oracle because they operate beyond the safe region. Equation (4) puts a price on that. Figure 3 plots the risk-adjusted profit relative to the oracle against the cost of off-spec bottoms. The fixed and CQR back-offs lose their advantage over certify-then-deploy once off-spec bottoms cost more than 0.032 and 0.030 USD/kg, and over sampling-and-discarding above 0.021 and 0.017 USD/kg. These thresholds are 2% to 4% of the 0.836 USD/kg value of the product, so any disposition of off-spec gasoline that costs more than a few percent of its value makes the guaranteed methods the better investment in expectation. The a posteriori plug-in loses to certify-then-deploy above 0.060 USD/kg, and sampling-and-discarding remains ahead of certify-then-deploy below 0.110 USD/kg.

The same figure carries a second lesson. At off-spec costs above about 0.064 USD/kg even the certified setpoint, which accepts $E(V) = 0.051$, earns more than the oracle designed for $V = 0.10$, and above about 0.077 USD/kg so does the conservative scenario design. The chance level α

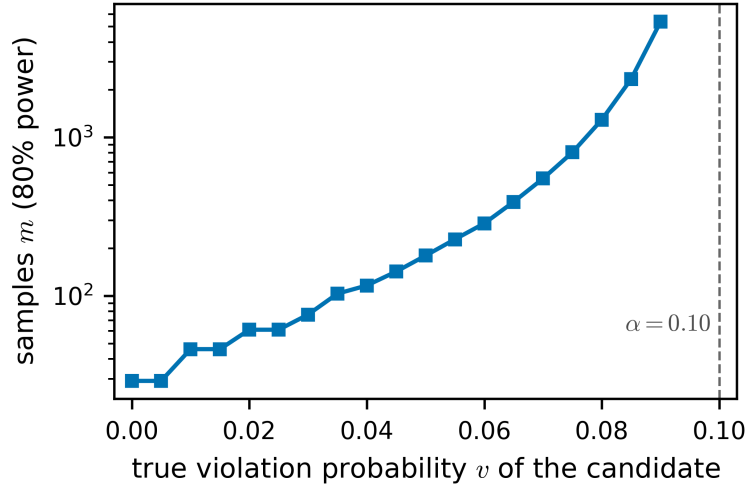


FIGURE 4 Evaluations needed to certify a candidate setpoint with 80% probability at $\alpha = 0.10$ and $\delta = 0.05$, against its true violation probability.

is therefore an economic decision rather than a convention: it should be set where the marginal profit of relaxing the constraint equals the expected cost of the extra off-spec production, and the procedures compared here can then enforce whatever level is chosen.

5.4 Certification budget and abstention

Certify-then-deploy needs enough samples to show that a candidate is safe. Figure 4 gives the number of evaluations that certifies a candidate with 80% probability as a function of its true violation probability. A candidate that never violates needs 29 evaluations; one at 0.02 needs 61, at 0.04 needs 116, at 0.06 needs 286, and at 0.08 needs 1290. The budget grows steeply as the candidate approaches α , so a finite budget always deploys a setpoint strictly inside the safe region. This is the price of the guarantee, and it is predictable before commissioning: an engineer can choose m from the figure and the profit gradient of the unit. When no candidate can be certified the method abstains, which on the twin happens increasingly for $\sigma_z \geq 0.020$, where the calibration data can no longer support a certified setpoint near the constraint. Abstaining is the correct outcome in that regime: it signals that the disturbance has outgrown the data.

6 SELECTION GUIDE AND LIMITATIONS

Table 2 condenses the evidence into a rule for choosing the back-off of a chance-constrained RTO.

The study has limits that bound these conclusions. The plant is a simulation twin, rigorous but not

TABLE 2 Choosing the back-off of a chance-constrained RTO.

Situation	Use	Evidence
A validated plant model can be evaluated inside the optimizer	Sampling-and-discarding	0.0% unsafe deployments, 3.2 USD/h below the oracle, no abstention for $\sigma_z \leq 0.015$
The optimizer uses a nominal or surrogate model; the plant model or pilot runs can be evaluated at a few candidate setpoints	Certify-then-deploy	At most 2.5% unsafe deployments at every disturbance level; budget from Figure 4; valid for any optimizer
Only a marginally valid conformal back-off or a plug-in correction is available	Do not deploy without certification	12.0% to 100% unsafe deployments across disturbance levels
The chance level has not been fixed	Set α from the off-spec cost	Guaranteed methods overtake the $\alpha = 0.10$ oracle above 0.064 to 0.077 USD/kg

an operating column, so the violation probabilities are relative to the twin, and the guarantees of certify-then-deploy and the scenario methods hold with respect to the model used to evaluate the candidates. The disturbance is a single scalar whose successive values are treated as independent; real feed compositions drift and are autocorrelated, which breaks the independence behind every guarantee compared here, and blocked or weighted certification is the natural next step. With several disturbances or a non-monotone response more scenarios would limit the solution, and the balance between sampling-and-discarding and certify-then-deploy may shift toward the latter, whose sample size does not grow with the number of decision variables. Finally, the economic thresholds depend on the price assumptions stated in Section 4 and scale with them.

7 CONCLUSIONS

Data-driven back-offs are a sound idea for RTO under unmeasured disturbances, but a back-off that is valid on average is not safe at the setpoint an optimizer chooses. On a rigorous debutanizer twin, conformal back-offs of this kind deployed a setpoint that exceeded the 10% violation target in 67.5% to 100% of trials, and a plug-in correction that is right on average still did so in 32.0%. Methods that check the deployed setpoint itself, certify-then-deploy and sampling-and-discarding, kept this rate at or below 2.5% for a profit cost of 0.06% to 0.21%, and they are the more profitable choice in expectation whenever off-spec bottoms cost more than 2% to 4% of their value. The practical rule is short: when the plant model can be evaluated inside the optimizer, use sampling-and-discarding; when only a nominal model can, certify the candidate setpoints before deploying them; and choose the chance level from the cost of off-spec product.

NOMENCLATURE

C	back-off added to the nominal model (—)
D	distillate molar flow rate (mol/s)
g	bottoms n -butane mole fraction, x_B (—)
$\hat{g}$	nominal model of g at z_0 (—)
$\bar{g}$	bottoms specification limit (—)
J	risk-adjusted profit rate (USD/s)
k	number of discarded scenarios (—)
m	certification sample size (—)
$\dot{m}_B$	bottoms mass flow rate (kg/s)
n	calibration sample size (—)
N	number of sampled scenarios (—)
N_j	violation count of candidate j (—)
R	reflux ratio (—)
U_j	Clopper-Pearson upper bound for candidate j (—)
u	decision vector, (R, D)
V	violation probability of a setpoint (—)
x_B	bottoms n -butane mole fraction (—)
z, z_0	feed n -butane mole fraction and its nominal value (—)
<i>Greek Letters</i>	
α	chance-constraint level (—)
δ	allowed probability of deploying an unsafe setpoint (—)
Δ	cost of off-spec bottoms per unit mass (USD/kg)
κ	back-off inflation factor (—)
π	operating profit rate (USD/s)
σ_z	standard deviation of the feed composition (—)

AUTHOR CONTRIBUTIONS

Bien Don Busico: Conceptualization; data curation; formal analysis; investigation; methodology; project administration; resources; software; validation; visualization; writing, original draft; writing, review and editing.

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CONFLICT OF INTEREST STATEMENT

The author declares no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in GitHub at <https://github.com/beebzy-droid/IPIS>, including the DWSIM campaign data (docs/module3/cjce/data), the scripts that regenerate every figure and table from fixed random seeds, and the evidence files.

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